



ORACLE

Oracle OS Management Hub

Next-generation OS management across distributed environments

Operation Advisory Team

Office of CTO, EMEA Technology Engineering

February, 2026



Requirements and challenges of infrastructure management



Deployment and provisioning

- Provisioning across on-premises and multicloud
- Automating VMs and Container deployments
- Standardize deployments, avoid configuration drift
- Meeting high availability requirements



Software management and updates

- Managing software inventory
- Updating and patching the infrastructure
- Managing security and vulnerability



Reporting, security, and compliance

- Reporting security compliance status
- Meeting Service level agreements (SLAs)
- Protecting against cyberattacks and security breaches
- Meeting regulatory requirements



Monitoring and assessment

- Managing container and VM workloads
- Reporting problems and Issues
- Optimizing performance

Managing the OS - Challenges



Compliance, availability

Operating systems require constant and consistent administration and management to help ensure security compliance, high availability, and optimal performance



Labor intensive, inefficient processes

Manual administration and in-house management tools are labor intensive and require independent, specialized expertise



Complexity

OS patching can be error prone and complex to manage throughout lifecycle environments, especially for larger deployments that need to scale across distributed environments



Need to improve productivity

Companies are looking to improve productivity and operate more efficiently and would rather have people spend time on tasks more important to their business and at hand

Oracle OS Management Hub

Management as a Service is the future, driven by and for distributed/multicloud environments

Simplified management

- Management and monitoring of updates and patches for OS environments

Distributed environments

- On-premises (private datacenter)
- Oracle Cloud Infrastructure (OCI):
 - Commercial, Government Cloud regions
- 3rd party cloud

Centralized control

- Scalable management of OS deployments through an intuitive console in OCI, CLI, and API with SDKs

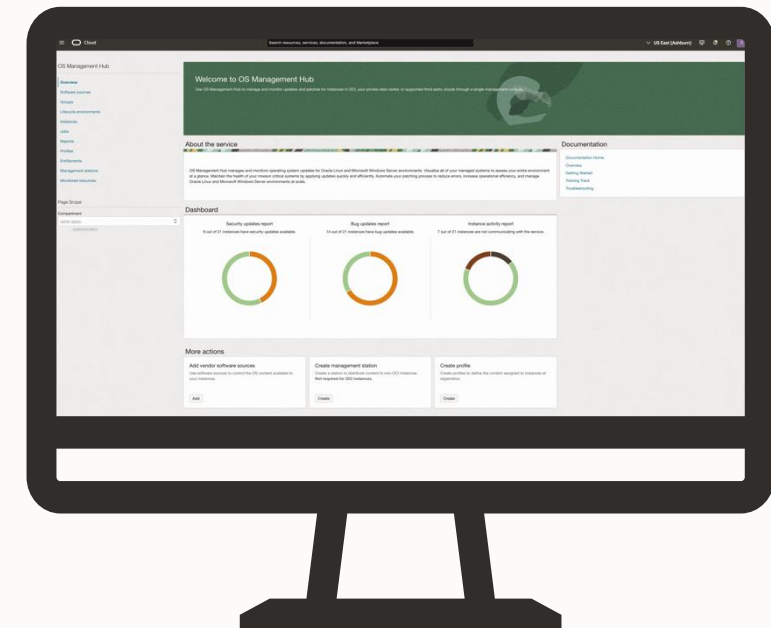
On-premises



OCI



3rd party cloud



Supported OS platforms

On-premises (Private datacenter)



- Oracle Linux 7
- Oracle Linux 8
- Oracle Linux 9

Oracle Cloud Infrastructure



- Oracle Linux 7
- Oracle Linux 8
- Oracle Linux 9
- Microsoft Windows Server 2016*
- Microsoft Windows Server 2019*
- Microsoft Windows Server 2022*

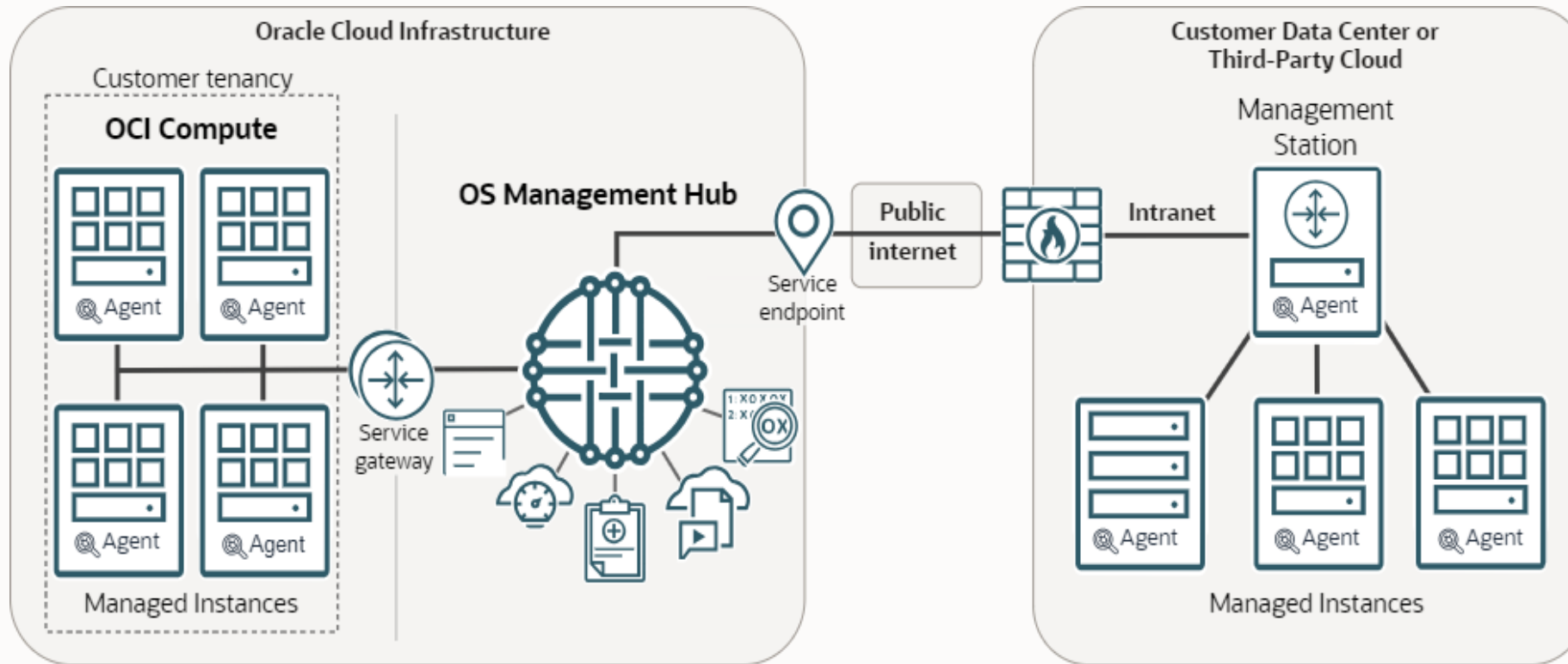
*Standard, Datacenter

3rd party cloud



- Oracle Linux 7
- Oracle Linux 8
- Oracle Linux 9

Highly available and scalable service architecture



Management station(s) communicate with the service

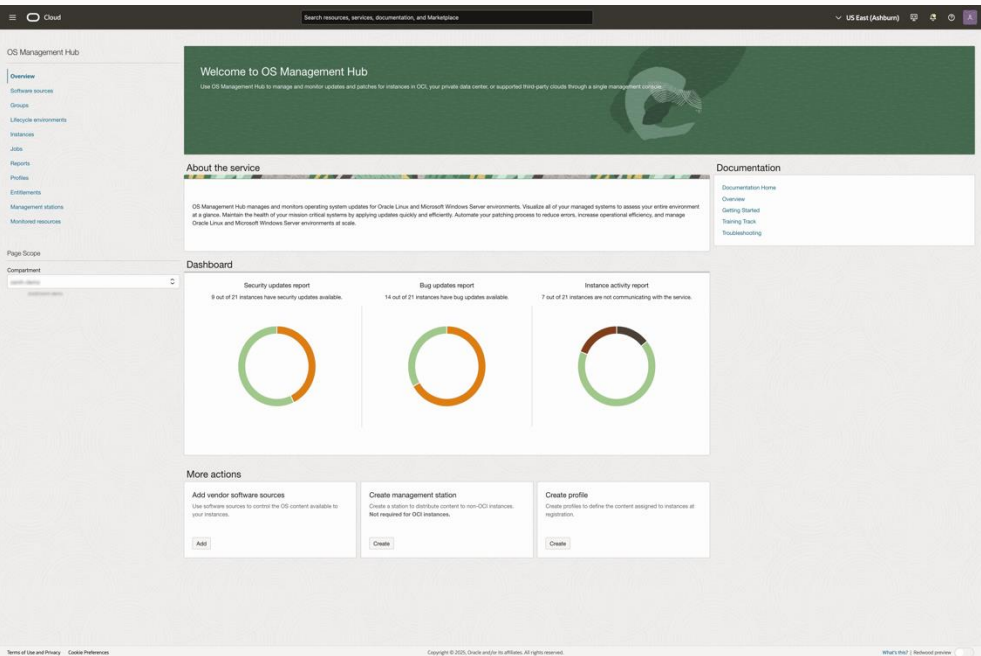
- Exists on your on-premises data centers or public cloud infrastructure (except OCI)
- Acts as a proxy to managed instances
- Controls interactions with the service and mirrors OS content locally, reducing amount of bandwidth used for patching instances

Monitor security compliance

Overview dashboard

Keep an eye on your managed environment:

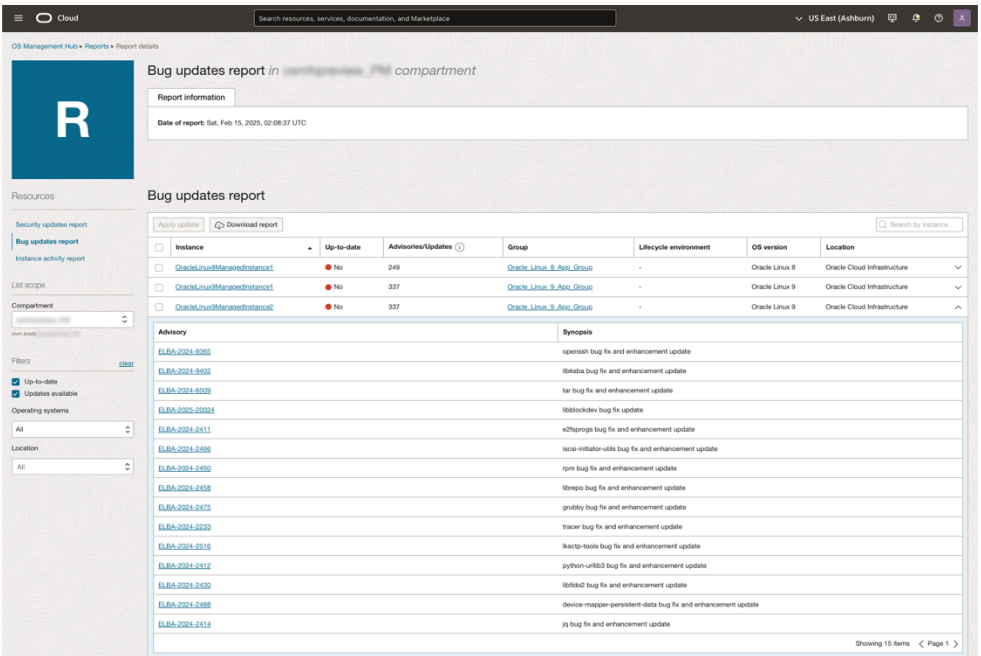
- Out-of-compliance patching statuses
- Number of available security and bug updates
- Instances communicating with the service



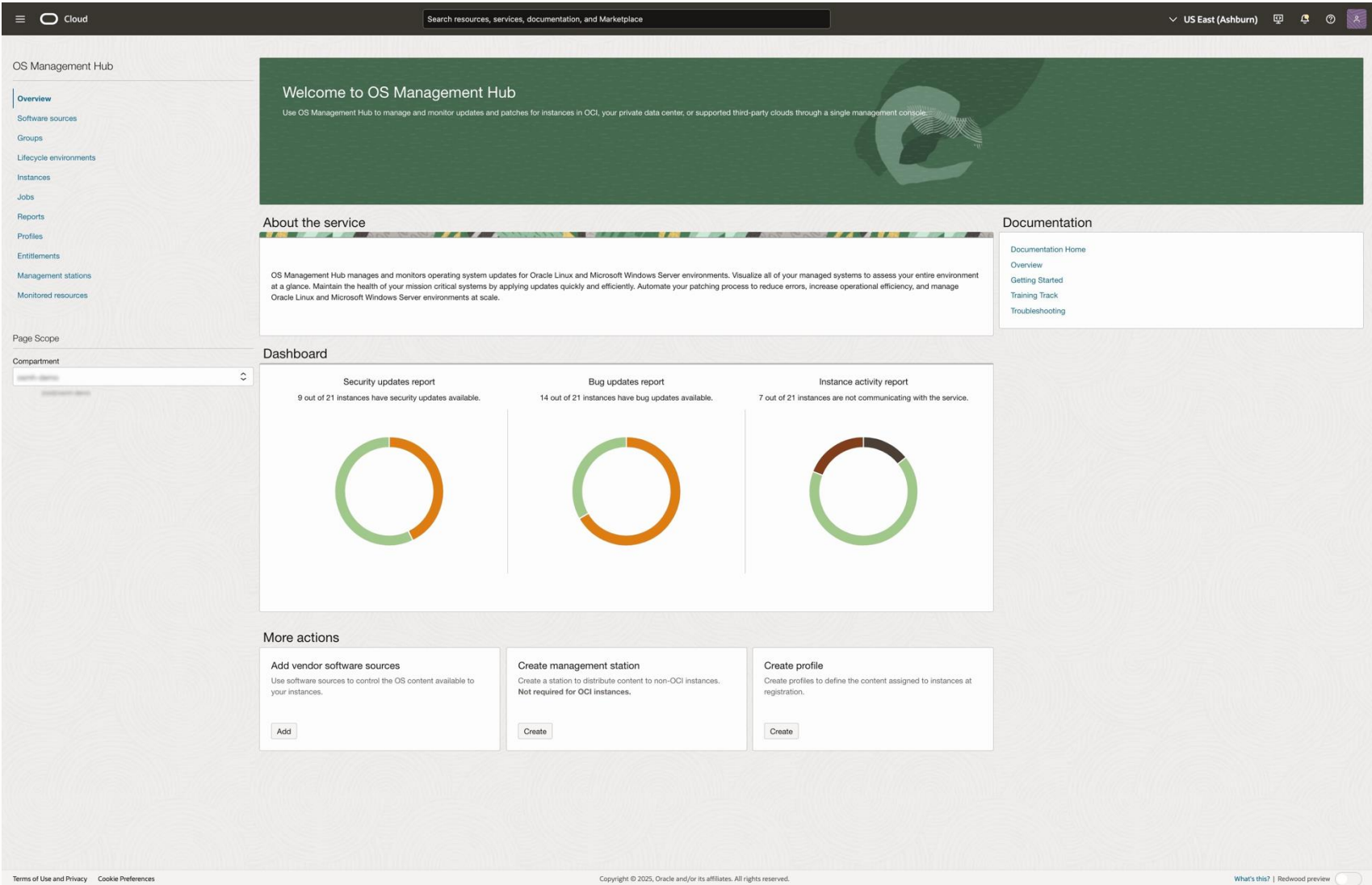
Real-time reports

Monitor whether an instance is up-to-date:

- Drill into linked security advisory information
- Run available updates directly from the Reports view
- Download reports in CSV, JSON, or XML formats

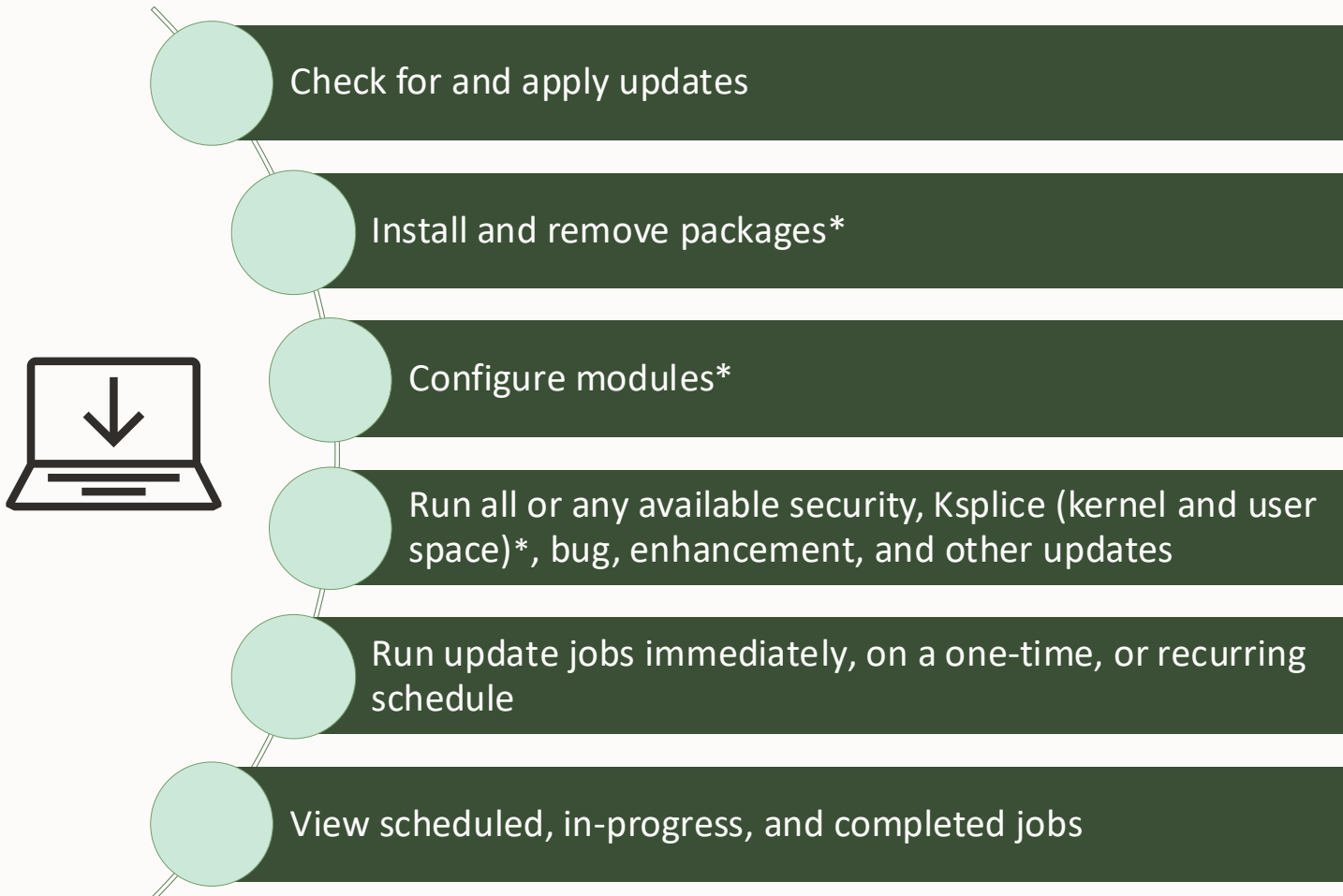


Overview dashboard



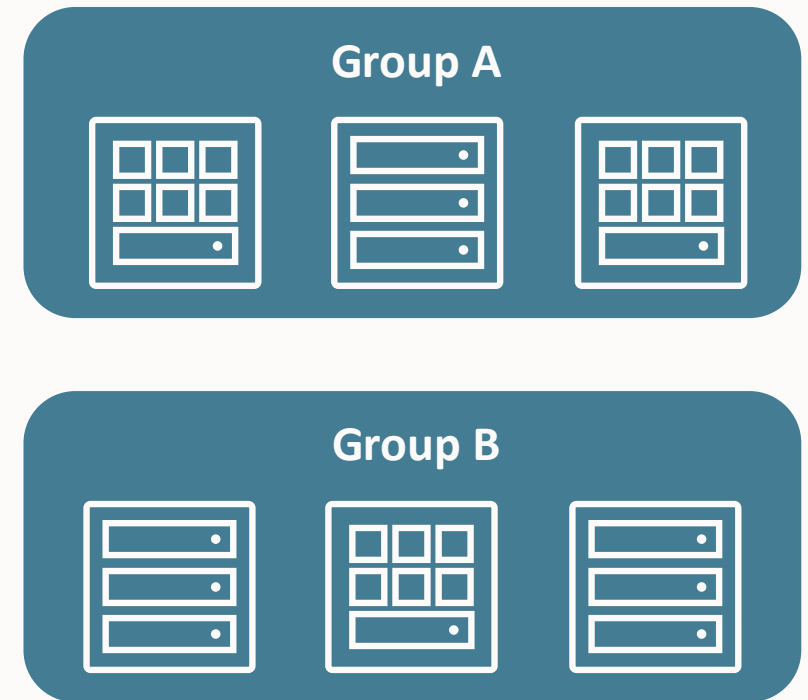
Simplify and automate patch updates

Efficient instance and group management through a single console



*Applicable only to Oracle Linux systems

Align OS content across a collection of instances
Perform fleet management operations



Why OS Management Hub?



Improve security and reliability

- Stay up to date with security and bug fixes
- Helps ensure security compliance



Save time, increase efficiency

- Automate manual processes; flexible job scheduling
- Automated operations reduce human errors



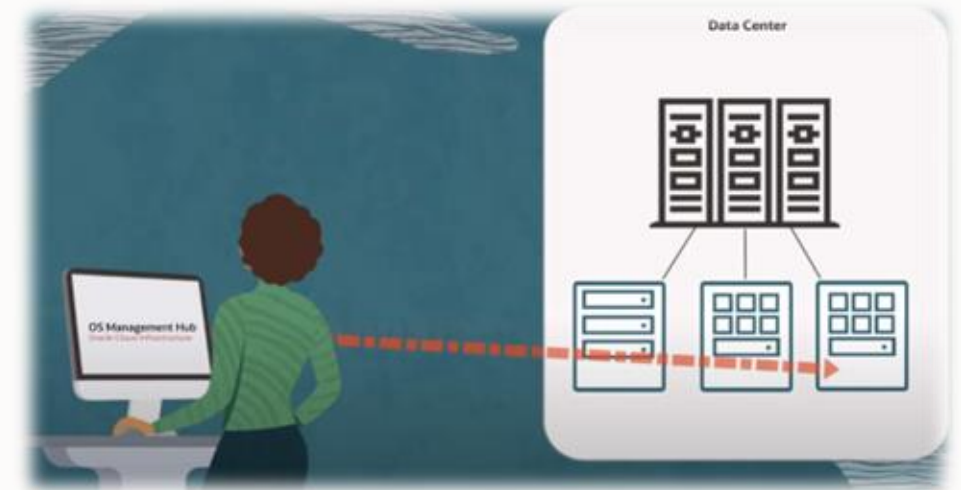
Reduce management complexity

- Centralized, simplified, integrated OS management
- Maintain consistency of patch lifecycle environments



Lower costs

- Management as a service reduces administrative overhead, operational, and capital expenditures



OS Management Hub entitlements

OS Management Hub entitlement	OCI Compute subscription
Manage Oracle Linux and Microsoft Windows instances in OCI	✓

OS Management Hub entitlement	Oracle Linux Basic Support	Oracle Linux Premier Support
Manage on-premises and 3rd party cloud Oracle Linux instances	✓	✓

Single *Oracle Linux Premier Support* SKU includes additional feature-rich infrastructure management solutions and value beyond technical support, such as:

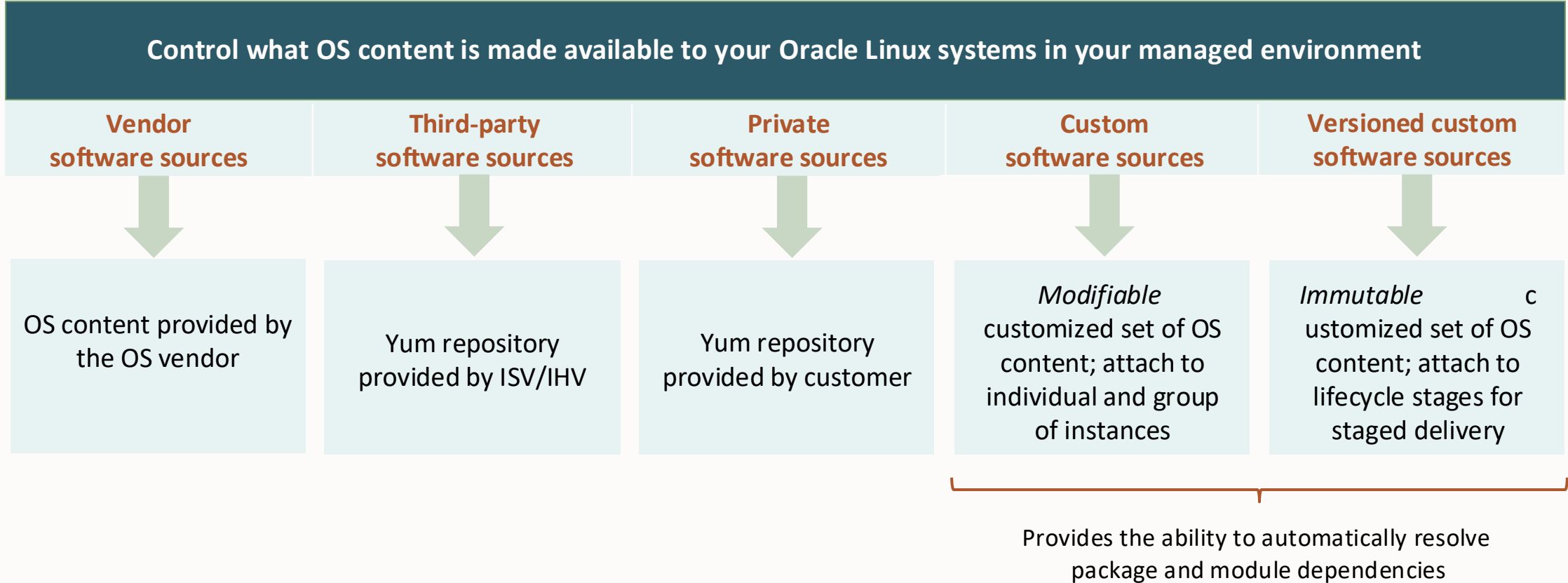
- Oracle Ksplice
- Oracle Linux Virtualization Manager
- Oracle Linux Automation Manager
- Oracle Cloud Native Environment



Managing Oracle Linux systems across distributed environments

Centralized control of OS content for Oracle Linux systems

Flexible management with the ability to customize



Control OS content from Oracle and non-Oracle software sources

Cloud

Search resources, services, documentation, and Marketplace

US East (Ashburn)

Create custom software source

1 Add basic details

2 Select software sources

3 Select package content

4 Review

Select vendor software sources.

OS vendor

Oracle

OS version

Oracle Linux 8

Architecture

x86_64

Available software sources

Compartment

own (prod/)

☒	Name	Description
☒	oel8_addons-x86_64	Oracle Linux 8 Addons (x86_64)
☒	oel8_appstream-x86_64	Oracle Linux 8 Application Stream Packages (x86_64)
☒	oel8_baseos_latest-x86_64	Oracle Linux 8 BaseOS Latest (x86_64)
☒	oel8_uekr7-x86_64	Oracle Linux 8 UEK Release 7 (x86_64)

4 selected

Previous

Next

Cancel

Cloud

Search resources, services, documentation, and Marketplace

US East (Ashburn)

Add third-party software source

1 Add basic details

2 Advanced options

3 Review

Basic details

Name

InfluxData_Repository

Description

InfluxData products, including InfluxDB, Telegraf, Chronograf, and Kapacitor

Configuration

URL

https://repos.influxdata.com/stable/x86_64/main/

OS version

Oracle Linux 7

Architecture

x86_64

X

OS version

Oracle Linux 8

Architecture

x86_64

X

OS version

Oracle Linux 9

Architecture

x86_64

X

Add

Allow mirror to management stations

☒ Enabled

Security

Enable GPG checking

☒ Enabled

GPG key URI

https://repos.influxdata.com/influxdata-archive_compat.key

Allow self-signed certificates

☐ Disabled

Show advanced options

Next

Cancel



Managing and monitoring Oracle Linux instances

View instance details

Install and remove packages

Configure modules

View upcoming and completed jobs

View reports

Cloud

Search resources, services, documentation, and Marketplace

US East (Ashburn)

OS Management Hub > Instances > Instance details

oracle-linux-8-managed-instance

Create update job Unregister Reboot

Create recurring update jobs

ACTIVE

Instance information

Compartment: (root)/ Location: On premises Scheduled jobs: 0 Management station: OSMH Management Station CA Edit Registration profile: Oracle Linux 8 Profile Effective kernel: - Updates available: 613 Security, 744 Bug fix, 7 Enhancement, 5 Other Uptime: 23 days, 4 hours, 26 minutes Last check-in: Sat, Feb 15, 2025, 03:47:10 UTC OCID: ...nxzzu4ja

Updates

1370 Available updates 613 security, 744 bug fixes, 7 enhancements, 5 others

Update

Check for and apply updates

	Name	Available version	Installed version	Architecture	Type	Advisory	CVE
☒	audit-libs	3.1.2-1.el8	3.0.7-5.el8	x86_64	Bug fix	ELBA-2024-3173	-
☒	avahi-libs	0.7-21.el8_9.1	0.7-21.el8	x86_64	Security	ELSA-2023-7836	CVE-2023-38469, CVE-2023-38472, CVE-2023-38470, CVE-2021-3468, CVE-2023-38471, CVE-2023-38473
☐	avahi-libs	0.7-27.el8	0.7-21.el8	x86_64	Bug fix	ELBA-2024-3179	-
☐	avahi-libs	0.7-27.el8_10.1	0.7-21.el8	x86_64	Bug fix	ELBA-2024-6985	-
☐	bash	4.4.20-5.el8	4.4.20-4.el8_6	x86_64	Bug fix	ELBA-2024-3237	-
☐	bcache-tools	1.0.8-3.101.0.2.el8	1.0.8-3.101.0.1.el8	x86_64	Bug fix	ELBA-2023-13036	-
☐	bcache-tools	1.0.8-3.101.0.3.el8	1.0.8-3.101.0.1.el8	x86_64	Bug fix	ELBA-2024-12179	-

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Applying updates to Oracle Linux instances

OS Management Hub > Instances > Instance details

oracle-linux-8-managed-instance

Create update job

Unregister

Reboot

Instance information

Compartment: (root/)

Location: On premises

Scheduled jobs: 0

Management station: [OSMH Management Station CA Edit](#)

Registration profile: [Oracle Linux 8 Profile](#)

Profile version: 1

Agent version: 3.0.184.linux.x86_64

ACTIVE

Resources

Updates

Packages

Modules

Events

Software sources

Jobs

Reports

Filters

Type

All

Updates

1370 Available updates

613 security, 744 bug fixes, 7 enhancements

Update

☐	Name	Available version
☐	audit-libs	3.1.2-1.el8
☐	avahi-libs	0.7-21.el8_5
☐	avahi-libs	0.7-27.el8
☐	avahi-libs	0.7-27.el8_1
☐	bash	4.4.20-5.el8
☐	bcache-tools	1.0.8-3.101
☐	bcache-tools	1.0.8-3.101

Create update job

1 instance will be updated.

Job name

Apply updates to oracle-linux-8-managed-instance

Description

Optional

Updates to apply

☐ All

☒ Security

- Ksplice kernel
- Ksplice userspace

☐ Bug fix

☐ Enhancement

☐ Other

Schedule

☐ Run immediately

☒ Schedule

Date and time

Apr 7, 2025 06:00 UTC

Frequency

☐ Once

☐ Hour

☐ Day

☐ Week

☒ Month

Interval (months)

1

Automatic retry

☒

Show advanced options

Create

Cancel

Jobs can be configured to apply specific updates:

- All available updates
- Security
 - Ksplice kernel and userspace
- Bug fix
- Enhancement
- Other

Flexible update job scheduling:

- One time
- Recurring (hour, day, week, month)



Updating non-Oracle OS content on Oracle Linux instances

ACTIVE

oracle-linux-9-managed-instance

Create update job Unregister Reboot

Instance information

Compartment: root/

Location: Oracle Cloud Infrastructure

Compute instance: [oracle-linux-9-managed-instance](#)

Scheduled jobs: 0

Registration profile: [Oracle Linux 9 Profile OCI instances](#)

Profile version: 1

Agent version: 3.0.6-release-OSM4-3-0-linux.x86_64

OS version: Oracle Linux 9

Architecture: x86_64

Kernel: 5.15.0-302.167.6.el9uek.x86_64

Effective kernel: 5.15.0-302.167.6.1.el9uek.x86_64

Updates available: [191 Security](#), [380 Bug fix](#), [7 Enhancement](#), [6 Other](#)

Uptime: 24 days, 9 hours, 50 minutes

Last check-in: Sat, Feb 15, 2025, 04:11:36 UTC

OCID: [lg6rh7q](#) [Show](#) [Copy](#)

Updates

584 Available updates

191 security, 380 bug fixes, 7 enhancements, 6 others

Update

☐	Name	Available version	Installed version	Architecture	Type
☒	chronograf	1.10.6-1	1.4.4.2-1	x86_64	@ Other
☒	influxdb2	2.7.11-1	2.0.3-1	x86_64	@ Other
☒	kapacitor	1.7.6-1	1.5.4-1	x86_64	@ Other
☐	oraclelinux-release	9.9.5-1.0.5.el9	9.9.4-1.0.6.el9	x86_64	@ Other
☐	oraclelinux-release-el9	1.0-18.el9	1.0-15.el9	x86_64	@ Other
☐	telegraf	1.33.2-1	1.10.0-1	x86_64	@ Other

3 selected

ACTIVE

oracle-linux-9-managed-instance

Create update job Unregister Reboot

Instance information

Compartment: root/

Location: Oracle Cloud Infrastructure

Compute instance: [oracle-linux-9-managed-instance](#)

Scheduled jobs: 0

Registration profile: [Oracle Linux 9 Profile OCI instances](#)

Profile version: 1

Agent version: 3.0.6-release-OSM4-3-0-linux.x86_64

Updates

584 Available updates

191 security, 380 bug fixes, 7 enhancements, 6 others

Update

☐	Name	Available version
☒	chronograf	1.10.6-1
☒	influxdb2	2.7.11-1
☒	kapacitor	1.7.6-1
☐	oraclelinux-release	9.9.5-1.0.5.el9
☐	oraclelinux-release-el9	1.0-18.el9
☐	telegraf	1.33.2-1

3 selected

Update package

3 updates will be installed.

Job name

Update 3 packages for oracle-linux-9-managed-instance

Description Optional

Schedule

☒ Run immediately ☐ Schedule

17

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Managing groups of Oracle Linux instances

View group details

Configure OS content

View upcoming and completed jobs

View reports

Cloud

Search resources, services, documentation, and Marketplace

US East (Ashburn)

OS Management Hub > Groups > Group details

Oracle_Linux_8_App_Group

Create update job Edit Create profile Move resource More actions

Group information

Tags

Compartment: (root/)

Description: Group for Oracle Linux 8 application servers in OCI

Associated instances: 2

Scheduled jobs: 0

OCID: ...6bjuowla Show Copy

OS version: Oracle Linux 8

Architecture: x86_64

Location: Oracle Cloud Infrastructure

Group members

Attach instance Detach instance Reboot

☐	Name	Compartment	Status
☐	oracle-linux-8-managed-instance-1	...	Active
☐	oracle-linux-8-managed-instance-2	...	Active

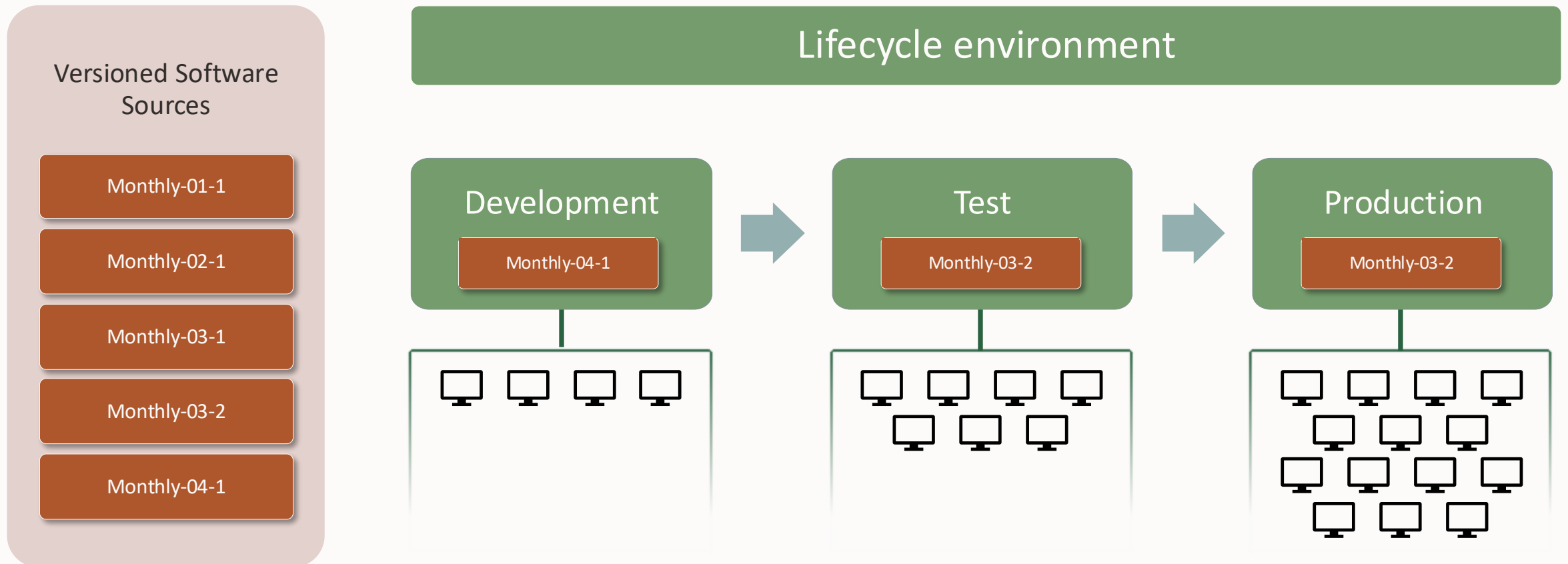
0 selected Showing 2 items Page 1

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Oracle Linux patch lifecycle management

Reduce management complexity — Ensure consistency and integrity of updates

A lifecycle environment is a user-defined pipeline to deliver curated, versioned content in a controlled manner



Lifecycle promotion of custom versioned software sources

Controlled delivery of Oracle Linux updates

LC

OS Management Hub > Lifecycle environments > Lifecycle environment details

DTP_Lifecycle_Environment

Edit

Create profile

Delete

Lifecycle information

Tags

Compartment:

OS version: Oracle Linux 8

Description: Lifecycle environment

Architecture: x86_64

Stages: 3

OCID: Show Copy

Instances: 3

Resources

Stages

Instances

Jobs

Reports

Stage number	Stage name	Versioned custom software source
☐ 1	Development	<a>o8-versioned 2.0
☐ 2	Test	<a>o8-versioned 2.0
☒ 3	Production	<a>o8-versioned 1.0

1 selected

Versioned custom software sources are created and promoted to the first stage, then subsequently promoted through the lifecycle stages

LC

OS Management Hub > Lifecycle environments > Lifecycle environment details

DTP_Lifecycle_Environment

Edit

Create profile

Delete

Lifecycle information

Tags

Compartment:

OS version: Oracle Linux 8

Description: Lifecycle environment

Architecture: x86_64

Stages: 3

OCID: Show Copy

Instances: 3

Resources

Stages

Instances

Jobs

Reports

Stage number	Stage name	Versioned custom software source
☐ 1	Development	<a>o8-versioned 2.0
☐ 2	Test	<a>o8-versioned 2.0
☒ 3	Production	<a>o8-versioned 1.0

1 selected

Promote content

The Production stage has a predecessor in the lifecycle environment. Content will be promoted from its predecessor.

Promoting o8-versioned from Test to Production

Job name

Promote o8-versioned to DTP_Lifecycle_Environment Production

Description Optional

Schedule

☐ Run immediately ☒ Schedule

Date and time

Jun 3, 2024 06:00 UTC

< June 2024 >

Su	Mo	Tu	We	Th	Fr	Sa
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

Time

04:30 UTC
05:00 UTC
05:30 UTC
06:00 UTC
06:30 UTC
07:00 UTC
07:30 UTC

Promote

Cancel

Promotion can be run immediately or scheduled at a certain date and time

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Vulnerability report for Oracle Linux instance

Cloud

Search resources, services, documentation, and Marketplace

US East (Ashburn)

OS Management Hub > Instances > Instance details

oracle-linux-8-managed-instance

Create update job Unregister Reboot

Instance information

Compartment: (root)

Location: On premises

Scheduled jobs: 0

Management station: [OSMH Management Station CA Edit](#)

Registration profile: [Oracle Linux 8 Profile](#)

Profile version: 1

Agent version: 3.0.184.linux.x86_64

OS version: Oracle Linux 8

Architecture: x86_64

Kernel: 5.15.0-200.131.27.el8uek.x86_64

Effective kernel: -

Updates available: [613 Security](#), [744 Bug fix](#), [7 Enhancement](#), [5 Other](#)

Uptime: 23 days, 5 hours, 59 minutes

Last check-in: Sat, Feb 15, 2025, 05:21:24 UTC

OCID: ...nxzzu4ja [Show](#) [Copy](#)

ACTIVE

Resources

Updates

Packages

Modules

Events

Software sources

Jobs

Reports

Reports

Vulnerability report Change history

Install update Download report

Search by advisory

Advisory	CVE	Description	Issue date	Affected packages
☒ ELSA-2024-8038	CVE-2024-34158 CVE-2024-34156 CVE-2024-34155 CVE-2023-45290	container-tools:ol8 security update	Mon, Oct 14, 2024, 00:00:00 UTC	15
☐ ELSA-2023-7549	CVE-2023-2163 CVE-2023-3812 CVE-2022-45884 CVE-2023-1192 CVE-2022-45919 CVE-2022-45886 CVE-2023-5178	kernel security and bug fix update	Fri, Dec 1, 2023, 00:00:00 UTC	8
☐ ELSA-2024-12582	CVE-2024-41091 CVE-2024-41090	Unbreakable Enterprise kernel security update	Mon, Aug 12, 2024, 00:00:00 UTC	4

Package	Version
bpftool	5.15.0-209.161.7.el8uek
kernel-uek	5.15.0-209.161.7.el8uek
kernel-uek-core	5.15.0-209.161.7.el8uek
kernel-uek-modules	5.15.0-209.161.7.el8uek

View outstanding advisories/security updates

Drill down on advisory information (see packages affected)

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Management Station Configuration

Management Station

Pre-requisites

Management station is mainly for **on-premises or third-party cloud Oracle Linux instances**. It mirrors software sources locally and also acts as a proxy back to OS Management Hub in OCI. OCI instances do **not** use management stations. Oracle's getting-started flow lists flow for creating MACS keys, creating the management station, registering it, then creating profiles and registering non-OCI instances.

Before creating the station, make sure the OS Management Hub service prerequisites are met:

- Correct IAM policies for the compartment
- Required software sources or entitlements
- **Management Agent Cloud Service install key** for non-OCI systems



Management Station Creation

System requirements

- VM / BareMetal in customer DC Oracle Linux 8 (minimum 8.7)
- Storage based on software sources and Linux versions of the instances to be managed
- Port tcp/443 for interaction with OSMH

Configuration

- Hostname or IP address (on the local network) of management station

Proxy Information (OSMH service)

- Permitted addresses
- Listening port
- Forward HTTP proxy (optional for on-prem network)

Mirror information

- Mount point
- Listening ports (HTTP and HTTPS)

Refer Documentation

<https://docs.oracle.com/en-us/iaas/osmh/doc/create-management-station.htm>



Management Station Registration, Instance setup & verification

Registration

- Install packages on management station (as management station functionality is implemented as Apache server application)
- Add station profile to the instance (The Management station that was created)
- Install the Management Agent on the instance (To be managed) that provides a service plugin to the Management Agent service
- Set up Management agent with Management agent install key (for same identity domain for authenticity)
- Verify Management Station Registration in OCI

Next step

Attach Software Sources , Sync Mirror, Create profiles, Register instances

Refer Documentation

<https://docs.oracle.com/en-us/iaas/osmh/doc/register-management-station.htm>



Management Station Details

Cloud

Search resources, services, documentation, and Marketplace

German

MgmtStation1

Edit

View profile

Move resource

Delete

Management station information

Configuration

Tags

Compartment: oci4

Description: -

OS version: -

Architecture: -

Software sources: -

Location: -

Mirror storage: -

Last check-in: -

Mirror sync schedu

Next mirror sync: -

Station managed in

Peer stations: - 24

OCID: ...2pvtzqqq \$

Instances receiving updates

Status

Jobs

Associated instances

Associated profiles

Registration profiles

Configuration details

Configuration

Tags

Mirror storage path: /mnt/bigfs/OSMHStorage

Mirror listening port (http): 30001

Mirror listening port (https): 30002

SSL certification path: -

SSL verification enabled: No

Status

Mirror sync status

Mirror volume

0% - Failed



Resources

OS Management Hub

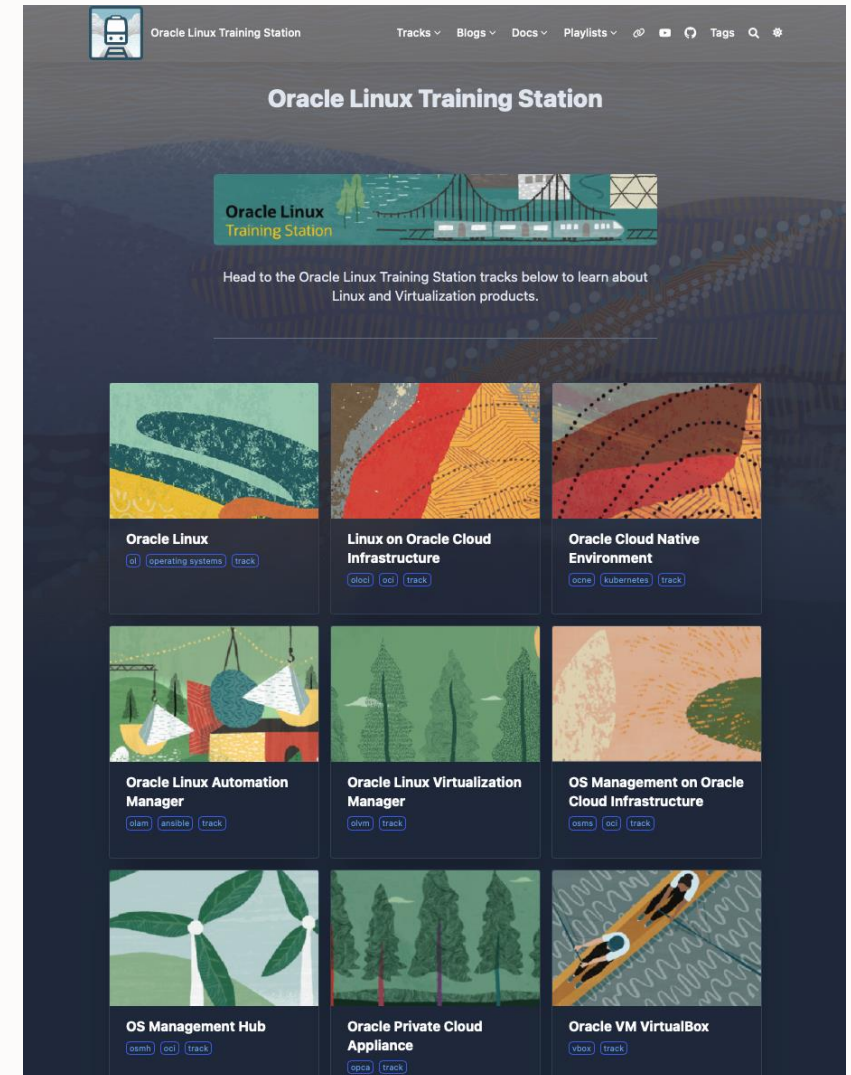
- <https://www.oracle.com/cloud/compute/os-management/>

Documentation

- OS Management Hub:
<https://docs.oracle.com/iaas/osmh/doc/home.htm>
- Oracle Linux documentation and training:
<https://docs.oracle.com/en/operating-systems/oracle-linux/>

Training/Videos

- Oracle Linux Training Station:
<https://www.oracle.com/goto/oltrain>
- OS Management Hub Learning Track:
<https://oracle-samples.github.io/oltrain/tracks/osmh/>



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blogs.oracle.com/linux

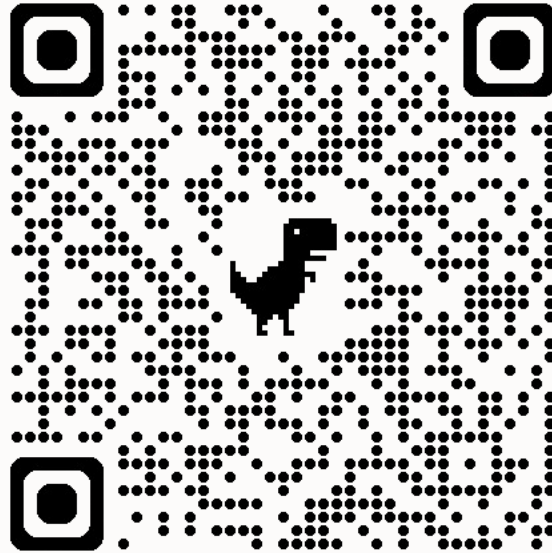


youtube.com/oraclelinuxchannel



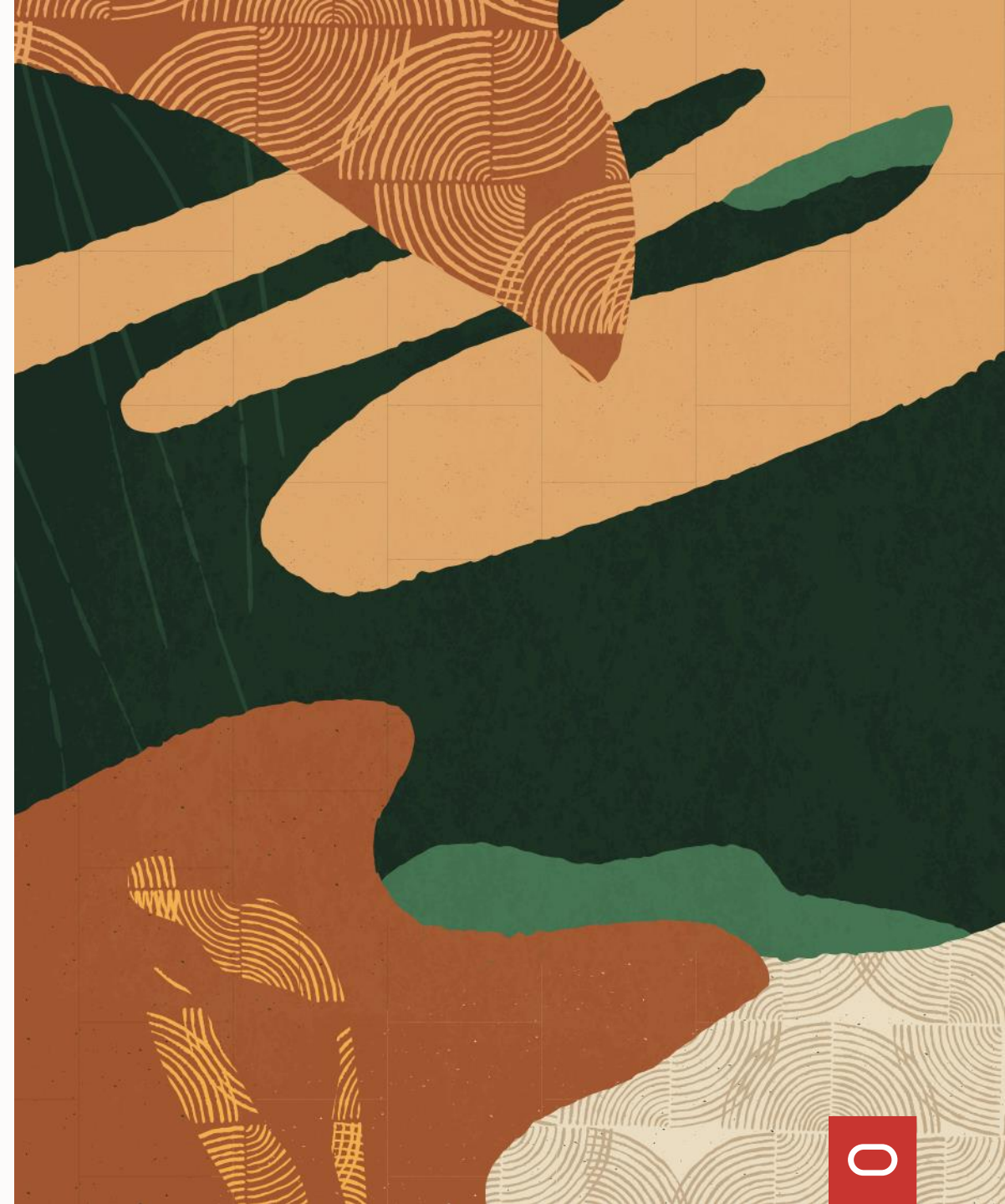
oracle.com/linux

Stay tuned



[Operations Advisory](#)
[GitHub Repository](#)

Thank you



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